COMPLETE RESEARCH GRANT PROPOSAL

Project SHIFT

1. Project Title

Analysis of the Subjective Horizon of Information (SHI) within Non-linear Determinism: Digital Ontology in Optimizing Perception for Biological Systems and AI (Project SHIFT).

2. Executive Summary (Abstract)

This project postulates the verification of the Subjective Horizon of Information (SHI) hypothesis, redefining human perception as a dynamically rendered, energy-efficient information system. Grounded in Friston's Free Energy Principle and Landauer's Principle, the project assumes that the physical parameters and coherence of matter undergo hard stabilization only within the radius of direct cognitive interaction (the "Operational Bubble" or BIP, approx. 1.5–6m). The research aims to measure the cognitive decoherence of objects outside this horizon and investigate the thermodynamic cost of resetting working memory. The findings will revolutionize cognitive ergonomics for critical system operators and data filtration architectures in autonomous vehicles (AVs), allowing for a 40-60% reduction in processor load.

3. State of the Art & Rationale

Current AI models and cognitive science paradigms suffer from a "combinatorial explosion"—they attempt to process the entire available environment at maximum resolution. The human brain avoids this through Interaction-Driven Culling. However, there is a lack of an empirical mathematical model describing the boundary (Horizon) beyond which the brain ceases to render objects. This project bridges this gap by integrating information physics with neuroacoustics, explicitly investigating how generative audio algorithms (e.g., AI, YouTube) modulate this rendering speed.

4. Research Hypotheses

- **H1 (Computational Irreducibility in Perception):** In repetitive spatiotemporal cycles, the accumulation of micro-variables leads to deterministic chaos, forcing the brain to constrict its Operational Bubble to conserve resources.
- **H2 (Spatiotemporal Decoherence of Objects):** Material objects that fall outside the direct attentional horizon undergo informational entropy, resulting in identification and localization anomalies (memory "glitches").

- **H3 (Multi-Observer Consensus as Error Correction):** The perceptual convergence of multiple agents in a shared space acts as an Error Correction Code, drastically reducing the probability of object decoherence.
- **H4 (Thermodynamics of Informational Reset):** Returning to optimal cognitive efficiency requires a quantified energetic expenditure to erase irrelevant variables (Selective Information Erasure / SHIFT-Landauer Principle: $\Delta S + \Delta I \geq k \ln 2$).

5. Methodology & Work Packages (WP)

- **WP1: Theoretical Modeling.** Developing equations for the SHI based on Shannon's information theory.
- **WP2: Behavioral Studies in VR.** Utilizing VR headsets with integrated eye-tracking to measure reaction times to objects outside the 1.5–6 meter radius.
- **WP3: Neuroimaging (fMRI / EEG).** Recording metabolic signatures during "variable overload" and mapping activity during the cognitive "System Reset" procedure under the influence of various algorithmic audio stimulants.
- **WP4: AI Implementation.** Translating findings into a Resource-Aware Perception Algorithm (Interaction-Driven Culling) for neural networks.

6. Proof of Concept & Validation

- **The Render Boundary Experiment:** Utilizing VR to prove that the cognitive system renders full physical parameters strictly within the BIP bubble (1.5-6m).
- **Verification of the "SHIFT-Landauer Principle":** Proving that erasing a bit of information requires a hard energy expenditure of $W = kT \ln 2$.
- **Cognitive Interferometry:** Confirming a measurable drop in informational density for objects located outside the R horizon.

7. Planned Clinical Trials

- **Neurorehabilitation & ADHD Treatment:** Managing the informational horizon of patients with attention deficits via a forced reset of variables in working memory.
- **Computational Psychiatry:** Modeling anxiety states as errors in the rendering of future variables.
- **Cognitive Hygiene & Burnout:** Administering the "Selective Information Erasure" protocol to restore thermodynamic homeostasis.
- **Surgical Ergonomics:** Optimizing the surgical environment based on the BIP bubble using algorithmically controlled AI audio to minimize errors arising from deterministic chaos.

8. Commercial & Business Impact

- **Autonomous Vehicles (AV):** Implementing Interaction-Driven Culling to reduce LiDAR noise, yielding a 25-30% increase in decision speed.
- **Logistics 4.0 & Time Loops:** Optimizing processes in 4-12h cycles, eliminating "route fatigue" and reducing human error by 15-22%, alongside a 15% drop in neural energy consumption.
- **Edge Computing (Deep Tech AI):** Decreasing power consumption in edge processors by up to 40%.
- **Remote Work / IT:** Applying acoustic attention stabilization systems (AI Music) to increase Deep Work efficiency by 40%.

9. Scientific & Medical Impact

- **The Digital Ontology Paradigm:** Redefining matter from a "physical constant" to a "dynamic information packet" dependent on interaction range (R).
- **Breakthroughs in Neuroacoustics:** Proving that AI-generated frequencies act as the brain's "clock speed," artificially expanding or constricting the SHI.